

Machine Learning

Logistic Regression- Week 7

Objectives

- Logistic Regression
- Cost function for Logistic Regression
- Reading Assignment (Andrew Ng , CS229 Lecture notes)

Slide credits:

Andrew Ng, CS229 Lecture notes

Where to find me?

Office:

Science Hall 250

Office Hours:

**Weds and Thurs 11:00 AM - 1:00 PM EST or by
appointment**

Logistic Regression

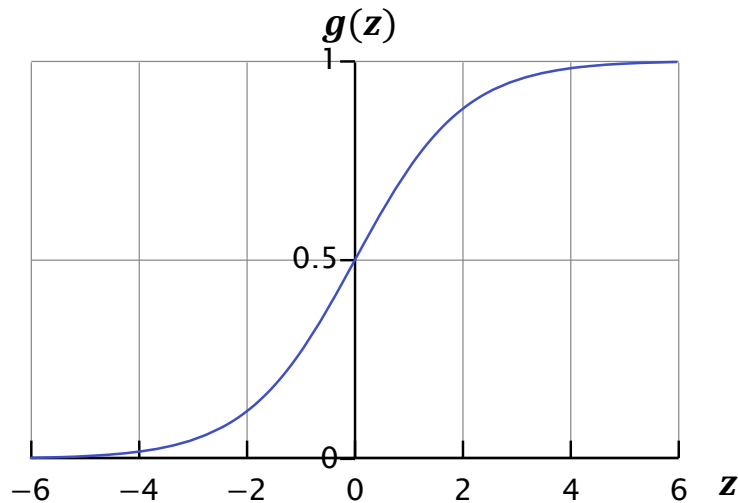
Want $0 \leq \hat{y} \leq 1$,

Linear regression equation : $f(x) = w^T x + b$

Now we define logistic regression as $f(x) = g(w^T x + b)$

Where $g(z) = \frac{1}{1 + e^{-z}}$

$g(z)$ is called the **sigmoid** or **logistic** function



Logistic Regression

Let's replace w by θ and b by θ_0

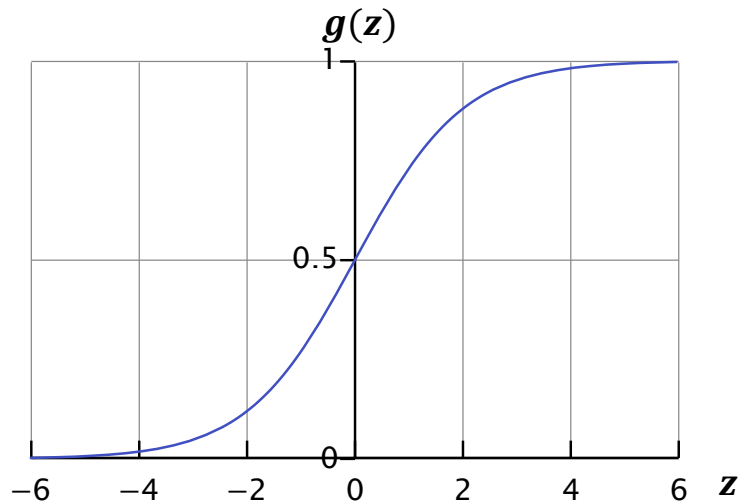
i.e. $f(x) = \theta_1 x + \theta_0 = \theta^T x$

Therefore, logistic regression as $f(x) = g(\theta^T x)$

Where $g(z) = \frac{1}{1 + e^{-z}}$

$$h_{\theta}(x) = f(x) = g(\theta^T x) = \frac{1}{1 + e^{-\theta^T x}}$$

$f(x)$ is often called hypothesis, $h_{\theta}(x)$ which is parameterized by θ



Interpretation of Logistic Regression

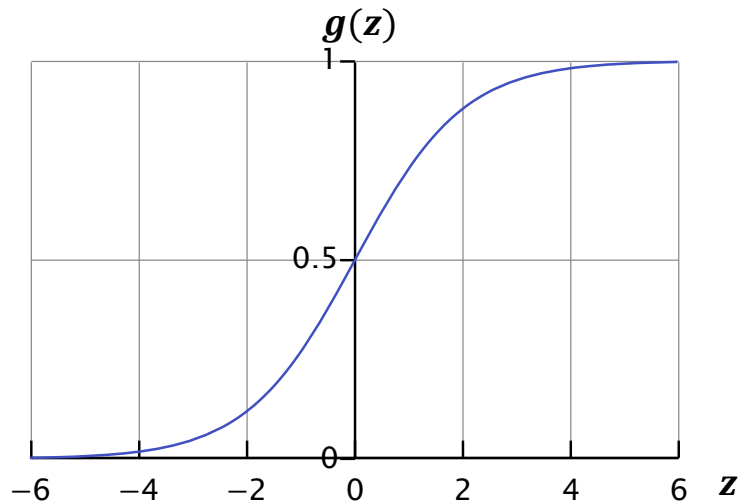
$$f(x) = \frac{1}{1 + e^{-\theta^T x}} = P(y = 1 | x, \theta)$$

Estimated probability that $y=1$ on input features x

Example: Let's go back to the tumor classification problem (malignant or not) based on size of the tumor (the feature we are using to classify a tumor)

If we use logistic regression as a classification model and it returns some value for a particular tumor size in test data:

$f(x) = 0.7 \rightarrow 70\%$ probability of tumor being malignant i.e. ($y = 1$)



Cost Function

For linear regression, we could either exactly solve the loss function or use gradient descent to numerically approximate the parameter values that makes the loss function a minimum.

For logistic regression, we could attempt to do the same. However, the cost function (the error between the predicted outcomes for the sample data and the actual outcome) is much more complicated than the simple least square sum:

$$J(\theta) = -\frac{1}{m} \sum_{i=1}^m (y_i \log(h_{\theta}(x_i)) + (1 - y_i) \log(1 - h_{\theta}(x_i)))$$

We can then apply gradient descent on the expression, $J(\theta)$ for to find values of θ where the minimum of $J(\theta)$ is reached.

ANY QUESTIONS?